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Submitter: Eva

Purpose and Scope

The purpose of this assignment was to explore the functionality of the Raspberry Pi single-board computer through hands-on configuration of its operating system, multimedia components, and peripheral hardware. This report documents the activities performed by the team, the results obtained, obstacles encountered, research conducted, and the knowledge gained during the process.

The objective was to understand how hardware and software interact in a Linux-based embedded system environment and to develop practical skills in configuring devices such as audio output, camera input, and the Sense HAT module.

Bootling and Initial System Setup

Activities Performed

The team prepared the Raspberry Pi for operation by installing the operating system on a microSD card using the Raspberry Pi Imager tool. The proper device model and operating system were selected, and the OS image was written on the microSD card.

During setup, the lab tablet normally used for configuration was not powered. As a result, the team connected the necessary cables to a campus computer to complete the installation process. A Raspberry Pi keyboard was used as the primary input device during configuration.

After installation, the microSD card was inserted into the Raspberry Pi. The system was connected to a monitor, keyboard, and power supply, and the Raspberry Pi was powered on and allowed to boot into the operating system environment.

Results Obtained

The Raspberry Pi booted successfully and displayed the graphical desktop interface. Input and output peripherals were recognized by the system, and the terminal environment was accessible for command-line operations.

Obstacles Encountered and Solutions

Special attention was given when selecting the storage device during installation to avoid accidentally overwriting other drives on the system. All hardware connections were carefully checked before powering on the Raspberry Pi to ensure the system started correctly and functioned as expected.

Learning Points

- The Raspberry Pi cannot run until an operating system is installed on a microSD card, which acts as its main storage and startup device.
- Careful installation and configuration directly affect how smoothly the system runs and how easily peripherals can be connected.
- Embedded systems like the Raspberry Pi rely on both correct physical setup and proper software configuration, showing how closely hardware and software must work together for successful operation.

Figure 1: Raspberry Pi OS desktop after boot



Figure 2: Raspberry Pi hardware setup with connected peripherals



Audio Output Configuration Using JBL Speaker

Activities Performed

The team successfully set up an audio link by connecting a JBL speaker with the Raspberry Pi. By reconfiguring the system's primary audio routing, we achieved seamless playback functionality. Initial stress tests confirm the connection is stable and the output quality meets project requirements.

Results Obtained

The Raspberry Pi successfully executed continuous audio transmission with the JBL speaker. The connection remained stable throughout the testing phase, showing no signs of latency or involuntary disconnection. Testing confirmed that the operating system consistently recognized the speaker and maintained stable audio playback.

Obstacles Encountered and Solutions

Initial setup required minor troubleshooting due to a configuration mismatch. The issue was successfully mitigated through a systematic review of the audio environment and a targeted reconnection protocol. Following these adjustments, the system kept a 100% success rate for device recognition.

Learning Points

- Audio peripherals require proper pairing and configuration.
- Audio output devices must be selected manually in system settings.
- External devices expand the multimedia capabilities of the Raspberry Pi.

Displaying Video Images from the Camera

Purpose

This activity focused on configuring and testing the Raspberry Pi camera and verifying that video and audio hardware functioned correctly using the Cheese application.

Activities Performed

The team accessed the command line interface and used the `sudo` command to check whether the Cheese application was installed. The system confirmed that Cheese was already available.

To launch the application, the command `cheese` was entered in the terminal. Initially, the camera did not start automatically, which caused confusion about how to activate it. The team researched how to run the camera using the Cheese application.

Within the Cheese interface, the Preferences menu was opened and the camera device labeled `iC700` was selected. External speakers were also connected to test audio output.

Results Obtained

After selecting the correct device, the camera activated and displayed a live video feed. The application functioned without errors, and the speakers produced audio successfully. The Raspberry Pi was able to run both video and audio components at the same time.

Obstacles Encountered and Solutions

The camera did not activate automatically when the application was launched. Research indicated that Linux-based systems sometimes require manual device selection. After choosing the correct device, the camera functioned properly.

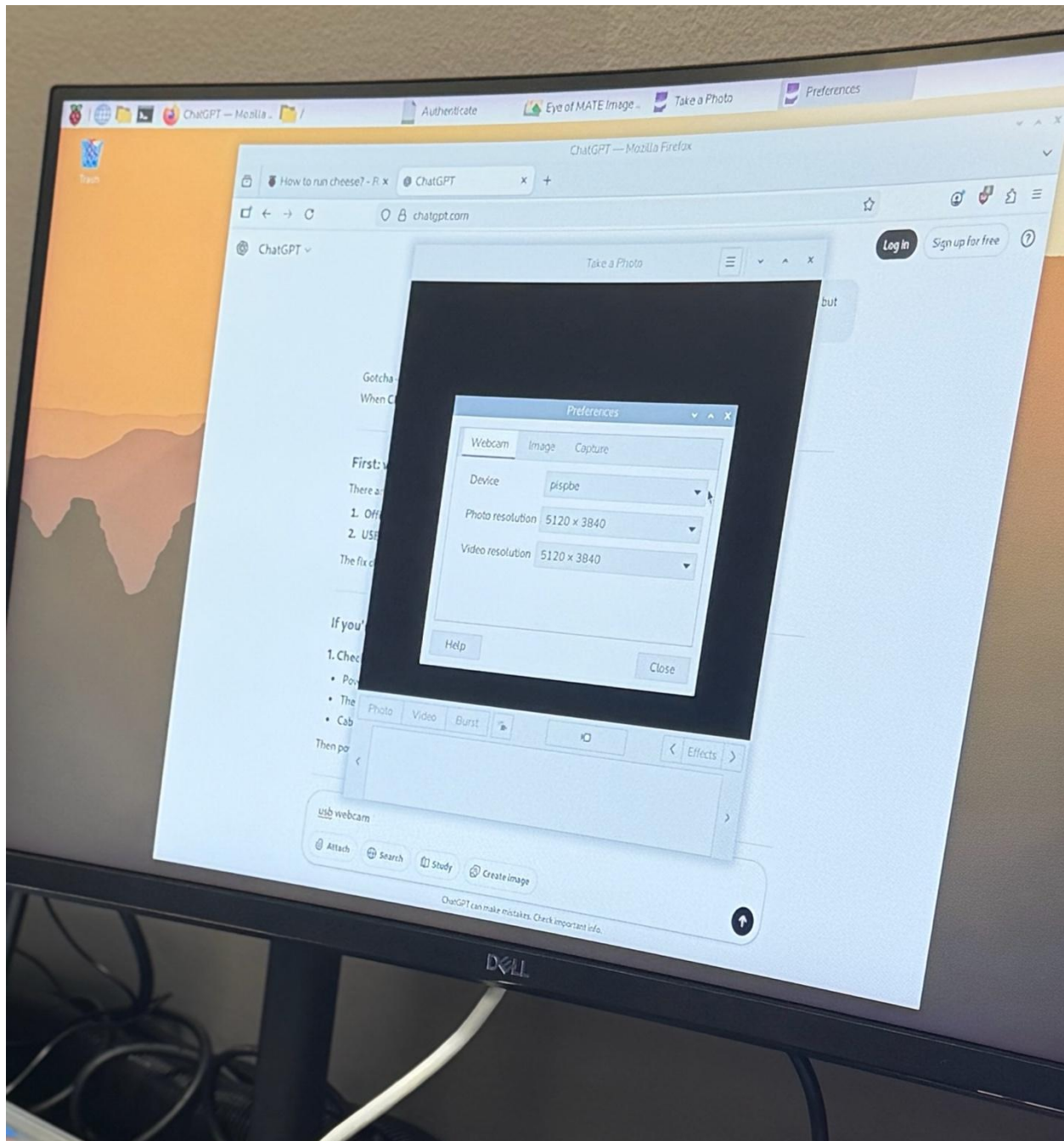
Research and New Concepts

The team researched how Linux applications detect and interact with hardware devices. Documentation explained that camera input devices may require manual configuration when automatic detection fails.

Learning Points

- Linux applications may require manual configuration to detect hardware devices.
- Research is an important step when troubleshooting unfamiliar system behavior.
- The Raspberry Pi can support simultaneous audio and video functionality when properly configured.

Figure 3: Camera device settings in the Cheese application



As shown in Figure 3, the camera device and resolution settings were configured within the Cheese application before activating the video feed.

Sense HAT Installation and Testing

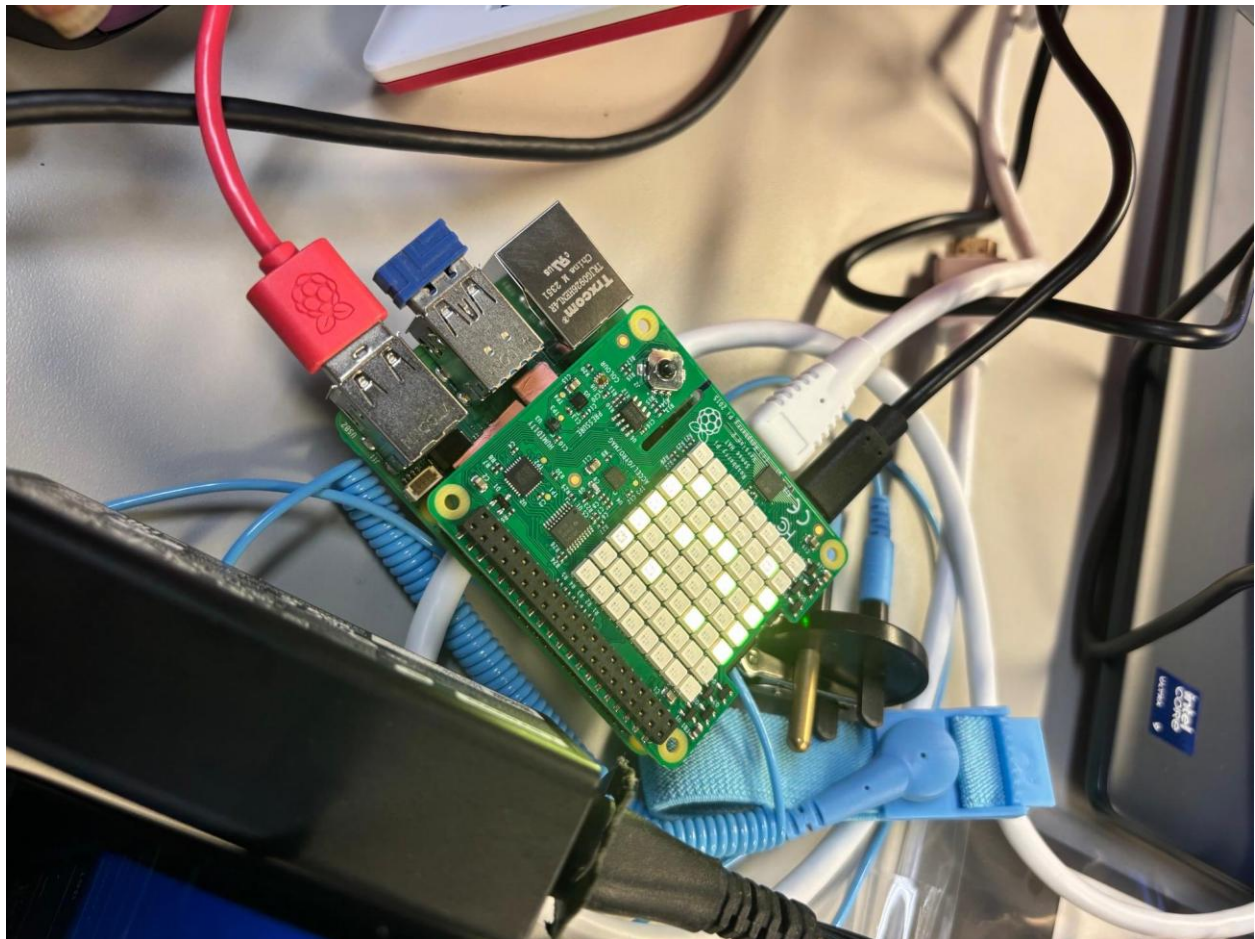
Purpose

The purpose of this activity was to install and test the Sense HAT module and verify its functionality through hardware setup and command-line interaction.

Activities Performed

The Sense HAT module was installed onto the Raspberry Pi to test its functionality. During the initial setup, the module was placed incorrectly, which prevented it from operating as expected. After examining the hardware configuration, the team repositioned the Sense HAT on top of the fan assembly to ensure proper alignment with the GPIO pins. Once the hardware was correctly installed, commands were executed in the terminal to initialize and test the Sense HAT interface. Throughout the process, system responses were monitored to confirm that the device was being recognized and configured properly.

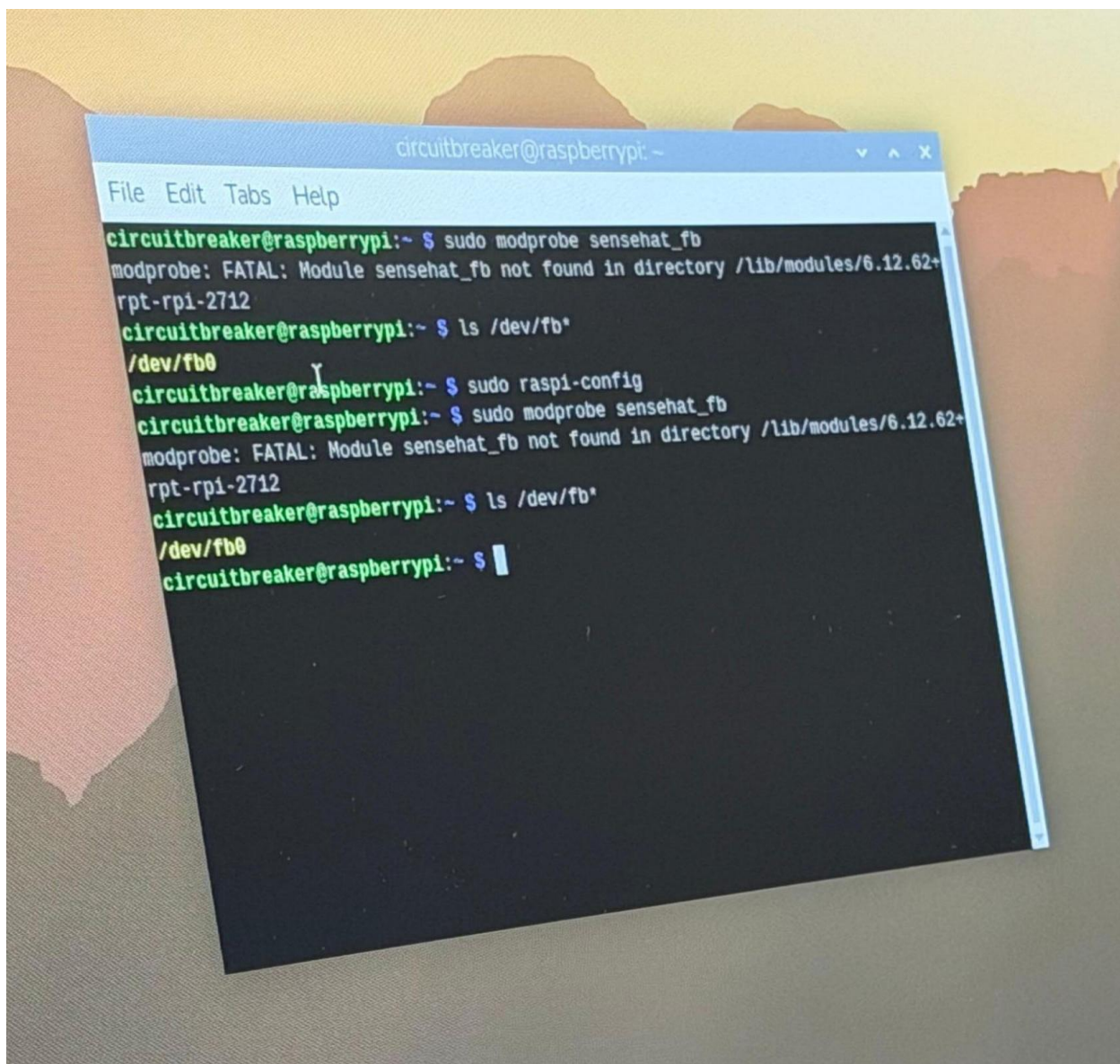
Figure 4: Correct Hardware Alignment and GPIO Pin Connection



Obstacles Encountered and Solutions

The Sense HAT did not function during the initial installation because it was placed incorrectly on the Raspberry Pi, which prevented proper alignment with the GPIO pins. After repositioning the module correctly on top of the fan assembly, the system was able to recognize the hardware. While running commands in the terminal, the system displayed the path `/dev/fb0` instead of the expected `/dev/fb1`, indicating that the display interface was not mapped correctly. The team continued troubleshooting by checking the hardware placement and command execution until the correct configuration was reached and Sense HAT responded as expected.

Figure 5: Terminal Diagnostics and Display Interface Mapping Error



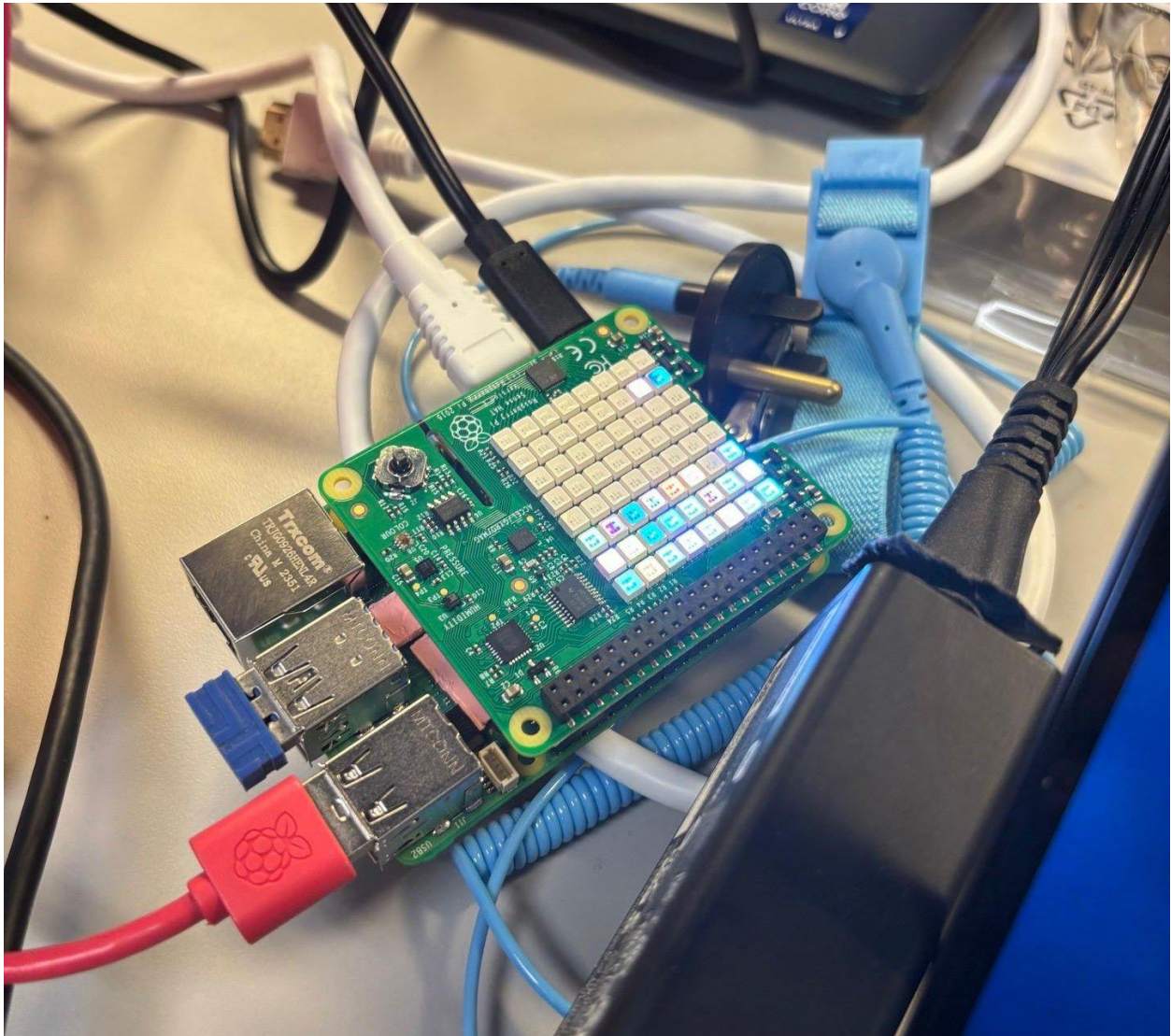
```
circuitbreaker@raspberrypi: ~
File Edit Tabs Help

circuitbreaker@raspberrypi:~ $ sudo modprobe sensehat_fb
modprobe: FATAL: Module sensehat_fb not found in directory /lib/modules/6.12.62+
rpt-rpi-2712
circuitbreaker@raspberrypi:~ $ ls /dev/fb*
/dev/fb0
circuitbreaker@raspberrypi:~ $ sudo raspi-config
circuitbreaker@raspberrypi:~ $ sudo modprobe sensehat_fb
modprobe: FATAL: Module sensehat_fb not found in directory /lib/modules/6.12.62+
rpt-rpi-2712
circuitbreaker@raspberrypi:~ $ ls /dev/fb*
/dev/fb0
circuitbreaker@raspberrypi:~ $
```

Results Obtained

After the Sense HAT was installed and positioned correctly on the Raspberry Pi, the system was able to detect the hardware without further issues. Once the appropriate terminal commands were executed, the device began responding as expected and reached its operational state. This confirmed that the Raspberry Pi and Sense HAT were communicating properly and that the configuration process had been completed successfully.

Figure 6: Successful Sense HAT Initialization and LED Matrix Output



Learning Points

- Proper GPIO alignment is critical for hardware functionality.
- Physical installation errors can prevent device recognition.
- Linux device paths such as `/dev/fb0` and `/dev/fb1` correspond to different display interfaces.
- Troubleshooting requires both physical inspection and command-line diagnostics.